Tri-plane diagrams and Minimum Crossing Number of Knotted Surfaces

XML Project 6
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Introduction

Definition (Triplane Diagram)

A **tri-plane diagram** is a triple of trivial tangle diagrams $P = (P_1, P_2, P_3)$ such that $P_i \cup \overline{P_j}$ is a diagram for an unlink. The **bridge number** of P is the number of strands in each of the three tangles. The **patch numbers** p_{ij} are the number of components in the unlink determined by $P_i \cup \overline{P_j}$.

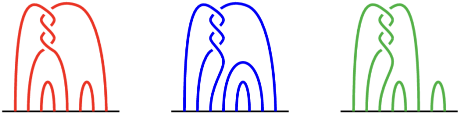


Fig 1. Example of Triplane Diagram.

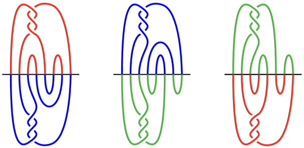


Fig 2. $P_i \cup \overline{P_j}$ is an unlink.

Triplanes sufficiently encode the structure of any knotted surface.

Definitions

Definition (Triple point)

A triple point is a triple intersection in a generic projection of a knotted surface (i.e. a point of intersection of 3 sheets).

Definition (Triple point number)

The triple point number $t(K)$ of a knotted surface K is the minimum number of triple points among all broken surface diagrams of K.

Definition (Crossing Number)

If K is a knotted surface, the crossing number is the total number of crossings in a tri-plane diagram of K, minimized over all diagrams.

Definitions

Definition (RIII Number)

The nth RIII number is given by

$$RIII(n) = max\{r(D) : \forall D \text{ with } n \text{ crossings}\}$$

where $r(D)$ is the minimum number of type three Reidemeister moves to transfer D into an unlink diagram.

Known Results

Theorem (J.Joseph, J.Meier, M.Miller, A.Zupan '22)

The number of triple points in the broken surface diagram of a knotted surface represented by a tri-plane diagram is the number of RIII moves required to simplify all of its pairwise unlink diagrams.

Theorem (S.Gong, S.Lewis-Monkman, J.Osnes '26)

Every 2-knot embedded in S^4 that admits a bridge trisection with at most five crossings is ribbon.

Theorem (S.Gong, S.Lewis-Monkman, J.Osnes '26)

A 5-crossing unlink requires at most 1 RIII to simplify to a 0-crossing unlink.

Theorem (Satoh '01)

For any positive n, there exists a surface-knot whose triple point number is 2n.

Known Results

Theorem (Triple point numbers for selected twist-spun knots. Satoh, Satoh-Shima)

Notation	*Knotted surface*	*Triple point number*
$\tau_2(3_1)$	*2-twist spun trefoil*	$t(\tau_2(3_1)) = 4$
$\tau_3(3_1)$	*3-twist spun trefoil*	$t(\tau_3(3_1)) = 6$
$\tau_2(4_1)$	*2-twist spun figure-eight*	$t(\tau_2(4_1)) = 8$
$\tau_2(T_{2,5})$	*2-twist spun* $(2,5)-$*torus knot*	$t(\tau_2(T_{2,5})) = 8$

Theorem (S.Gong, S.Lewis-Monkman, J.Osnes '26)

$\tau_2(3_1)$ *has 6 crossings.*

Main Results and Corollaries

Theorem

For any knotted surface K, if $t(K) > 4$, then $c(K) > 6$; if $t(K) > 6$ then $c(K) > 7$.

Corollary

The lower bound on $c(\tau_2(4_1))$ is 8.

Theorem

$c(\tau_2(4_1)) = 8$.

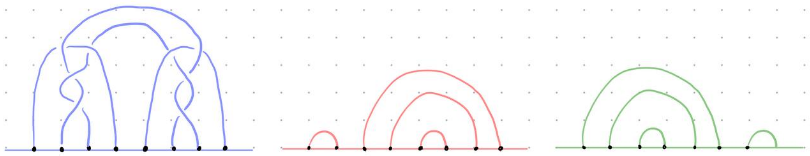


Figure 1: 8-crossing diagram for $\tau_2(4_1)$ (Fox)

Main Results and Corollaries

Corollary (Bounds on crossing numbers)

$14 \geq c(\tau_2(T_{2,5})) \geq 8$
$11 \geq c(\tau_3(3_1)) \geq 7.$

Corollary (Known and Conjectured $RIII(n)$ for n crossings)

Crossing number, n	*1*	*2*	*3*	*4*	*5*	*6*	*7*	*8*	*9*	*10*
RIII(n)	*0*	*0*	*0*	*0*	*1*	$2^{\dagger}$	$3^{\dagger}$	*4**	*5**	*?*

Table 1: *RIII(n), † denotes manually verified new result, * denotes computer aided result, not manually verified yet.* $RIII(5) = 1$ *from Thm 8.*

Process: Planar 4 valent graph with no self loops.

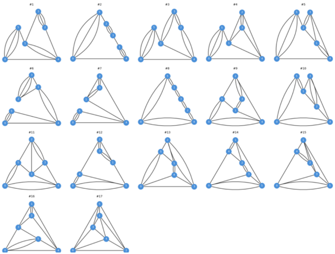


Figure 1: 6-vertex 4-valent graphs.

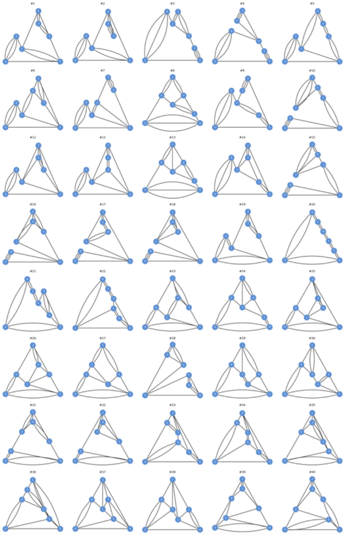


Figure 2: 7-vertex 4-valent graphs.

Process: Filter for unlinks and required RIII moves from all possible resolutions of 4-valence graphs.

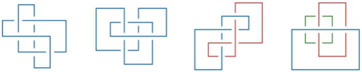


Figure 1: List of all 6 crossing unlink diagrams requiring at least one RIII moves to simplify them.

After enumerating the list of 6,7,8, and 9 vertex 4-valent graphs we determined the list of all n-crossing unlink diagrams. We determined the maximum number of RIII moves required to simplify them

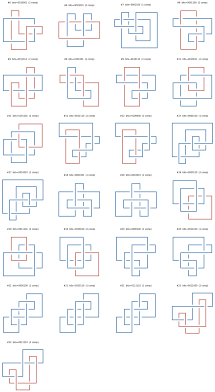


Figure 2: List of all 7 crossing unlink diagrams requiring at least one RIII moves to simplify them

Process: Manually verify R moves for simplifying 7-crossing unlinks.

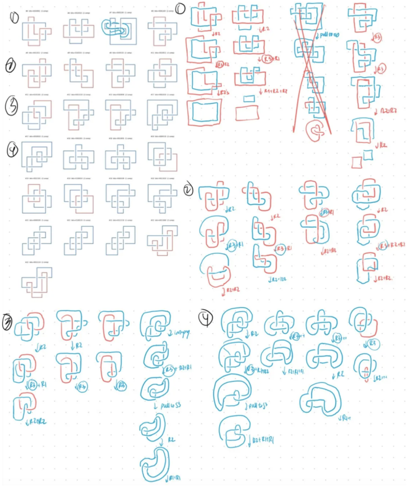


Figure 1: Simplification process for valid 7-crossing links.

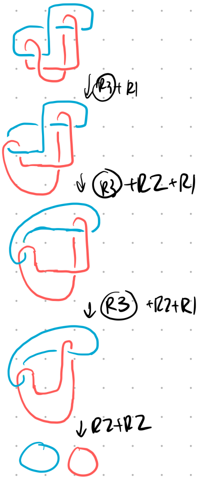
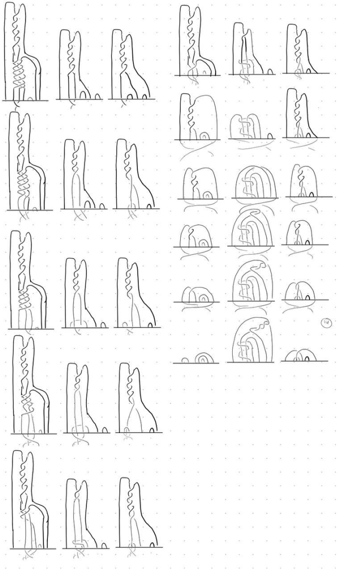


Figure 2: One case required 3 RIII moves.

Process: Reduction of $\tau_2(T_{2,5})$ to 14 crossings



Ribbon Knots

We also prove some results for ribbon knots, and find lower bounds on tri-plane crossing number for ribbon knots.

Lemma

Every 4-crossing unlink diagram can be simplified monotonically using only RI, RII moves. (Proof by hand)

Lemma

If D_C is the number of double circles in the broken surface diagram $\mathbb{K}$ of a 2-knot K obtained by the untying sequence of a tri-plane diagram (as in J.Joseph, J.Meier, M.Miller, A.Zupan). Then $D_C \leq \frac{1}{2}c(K)$

Ribbon Knots (Contd.)

Lemma

If $\mathbb{K}$ is a broken surface diagram of a 2-knot K which is free of triple points with $D_c \leq 1$ then $\pi_1(S_4/K) \cong \mathbb{Z}$

Theorem

Every 2-knot admitting a tri-plane diagram with at most 3 crossings is unknotted

Corollary

$4 \leq c(\tau_1(3_1)) \leq 6$

Current Explorations: RIII Move

$RIII$ move provides relationship between $t(K)$ and $c(K)$:

$t(K) \to RIII(n) \to \min\{c(K)\}, c(K) \to \max\{r(K)\} \to \max\{t(K)\}$

Proposition

For any two knotted surface with crossing number $n, m \in \mathbb{Z}$, and $2-$knots K_n, K_m with n crossing and m crossing, we have:

$$RIII(n) + RIII(m) \leq RIII(n+m), r(K_n) + r(K_m) \leq r(K_n \# K_m)$$

Proposition

For any $n \in \mathbb{N}$, we have $RIII(n) + 1 \leq RIII(n+5)$

Corollary (Weak Monotone increasing)

There exists a subsequence of $(RIII(n))_{n\in\mathbb{N}}$ which is strictly increasing.

Interesting Observations

Proposition

There exist knotted surfaces with arbitrarily large crossing number. This follows directly from (Satoh '01) and the fact that it implies that for any n, there exists a knotted surface with triple point number $2 \cdot (3 \cdot RIII(n))$

Proposition

Any knotted surface K with $t(K) > 3(49c)^{11}$ has crossing number greater than c. This follows from a result of Lackenby('15) that the number of Reidemeister moves required to simplify an unlink diagram is at most $(49c)^{11}$. And such knotted surfaces are indeed realized (Satoh '01).

Further directions

Direction (1)

Finding upper bound on crossing number via banded unlink diagrams

Direction (2)

Generalize lower bounds to arbitrary n crossing number

Direction (3)

Determine further lower bounds

Vakul says:

Vakul says: MIAOW :3 UwU